

MOHAMMAD SEDIGH DAKKALI

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SUMMARY

Medical doctor and clinical researcher with 20+ peer-reviewed publications and experience in ophthalmology, neuro-ophthalmology, prediction modelling, machine learning, clinical data analysis, and study design. My research interests focus on clinical prediction models, risk stratification, and data-driven approaches to improve clinical decision-making. I am seeking a PhD position in clinical research, particularly in prediction modelling and prognostic research.

WORK EXPERIENCE

Research Assistant, Alzahra Eye Hospital, Zahedan, Iran

Nov 2021 - Present

- Assisted in the planning and conducting clinical research projects.
- Managed the complete lifecycle of research projects, from conceptualization and design through to data analysis and reporting.
- Collaborated with research teams to develop methodologies, collect data, and conduct detailed analyses to drive strategic insights.
- Contributed to the drafting, submission, revision, and publication of papers.

EDUCATION

Doctor of Medicine

Sep 2013 - May 2021

Iran University of Medical Sciences, Tehran, Iran

- GPA: 17.75/20

RESEARCH INTERESTS

- Clinical Research
- Prediction models
- Prognostic studies
- Public Health

SELECTED PUBLICATIONS

Arish M., Hashemzahi Z., Baghaei V., Maleki A., Salari A., Rasouli S., **Dakkali M.S.*** (2025). *MicroRNA Signatures in Alzheimer's Disease and Normal-Tension Glaucoma: A Comparative Expression Analysis of miR-128 and miR-455-3p*. Curr Alzheimer Res. 2026 Jan 6. doi: 10.2174/0115672050425207251129073017. Epub ahead of print. PMID: 41508961.

Rasouli, S., **Dakkali, M.S.**, Ghazvini, A., Azarbad, R., Asani, M., Mirzaasgari, Z., Arish, M. (2024). *Predictive model for converting optic neuritis to multiple sclerosis; decision tree in focus*. PloS one. <https://doi.org/10.1371/journal.pone.0309702>.

Arish M., Sargazi M., **Dakkali M.S.**, Mohammadzamani S., Rasouli S., Asani M. (2024). *Effect of brimonidine on visual indices in patients with acute optic neuritis: a single blind randomized clinical trial*. Multiple Sclerosis and Related Disorders. <https://doi.org/10.1016/j.msard.2024.105913>.

Rasouli S., **Dakkali M.S.**, Azarbad R., Ghazvini A., Asani M., Mirzaasgari Z., Arish M. (2024). *Predicting the Conversion from Clinically Isolated Syndrome to Multiple Sclerosis: An Explainable Machine Learning Approach*, Multiple Sclerosis and Related Disorders. <https://doi.org/10.1016/j.msard.2024.105614>.

SELECTED PUBLICATIONS

Arish, M., **Dakkali, M.S.** (2023). *Evaluating the application of argon laser on pterygium surgery: Report of 30 patients*. Oman Journal of Ophthalmology. https://doi.org/10.4103/ojo.ojo_59_23.

ADDITIONAL PUBLICATIONS

- Machine-learning models for Alzheimer's disease classification using routinely available clinical, cognitive, and neuropsychiatric variables. BMC Neurology (under review)
- Optic Neuritis and Risk of Conversion to Multiple Sclerosis: A Regional Retrospective Cohort from Iran. Multiple Sclerosis and Related Disorders. (under review)
- Combination therapies in spinal muscular atrophy: Systematic review, 2025.
- Hypertension with/without LVH and microRNA expression, 2025.
- Phytosome nanocarriers for enhanced bioavailability, 2025.
- Sitagliptin, diabetes, and heart failure review, 2025.

SKILLS

Methodology:

clinical trials, cohort and case-control studies, prognostic modelling, prediction models, diagnostic accuracy research.

Tools:

Stata (regression, survival analysis, modelling)

R (machine learning: KNN, Naive Bayes, SVM, Random Forest, Decision Tree)

SPSS (clinical analysis)

COURSES

- Advanced Regression Methods with Stata (in progress)
- Machine Learning Course with R (2025)
- Designing, Analyzing & Writing Prognostic Value Studies (2025)
- Designing, Analyzing & Writing Prediction Model Studies (2024)
- Modern Methodology of Research (2024)
- Applied statistics using Stata and MedCalc (2024)
- Designing, Analyzing & Writing Diagnostic Test Accuracy Studies (2024)
- Designing, Analyzing & Writing Cross-Sectional Studies (2024)
- Systematic Review and Meta-analysis (2023)

AWARDS & HONORS

- Selected Researcher of Zahedan University of Medical Sciences (2022).
- Member, National Elite Foundation of Iran (since 2021).
- Ranked 5th in the Comprehensive Basic Medical Sciences Exam, Iran (2016).

REFERENCES

Mohammed Arish, MD

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Zahedan University of Medical Sciences
Email: Mohammad.erish@zaums.ac.ir

Saeid Rasouli, MD

Post Doctoral Fellowship
Department of Ophthalmology
John Hopkins School of Medicine
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